(a) Leave-one-excitation-out: 40 held-out contrasts (5 per state, set-ups in alphabetical order)
DS1
DS2
DS3
DS4
DS5
DS6
DS7
DS8
held-out intensity estimate (truth = 1)
2
1
0
−1
prior [-1, 1] = discrepancy-aware certificate
noise-only interval covers truth
noise-only interval misses truth
(b) Interval behaviour in transfer (320 queries)
fraction of held-out queries
1.0
0.8
0.6
0.4
0.2
0.0
naive
noise-only certificate
with discrepancy
0.23
0.47
1.00
0.60
0.50
0.00
0.26
0.23
0.00
covers true intensity
excludes 0 for imposed state
excludes 0 for a non-imposed state
(c) Direction-only attribution: 52% (chance 12.5%)
imposed state
1 2 3 4 5 6 7 8
nearest training direction